

Hao Yu

Postdoctoral Research Associate

Notre Dame Radiation Laboratory, University of Notre Dame

hyu5@nd.edu

ORCID 0000-0002-8574-701X

APPOINTMENTS

Postdoctoral Research Associate

University of Notre Dame, USA

PI: Prof. Sylwia Ptasińska

Feb 2025 – Present

Postdoctoral Researcher

National Institutes for Quantum Science and Technology (QST), Japan

PI: Prof. Yasunari Maekawa

May 2022 – Jan 2025

Academic Support Staff / Academic Specialist

The University of Tokyo, Japan

PI: Prof. Shinichi Yamashita

Apr 2020 – Apr 2022

Cooperative Researcher

National Institutes for Quantum Science and Technology (QST)

Apr 2020 – Mar 2022

Research Fellowship for Young Scientists (DC2)

Japan Society for the Promotion of Science (JSPS)

Apr 2018 – Mar 2020

Research Assistant (SEUT-RA)

The University of Tokyo, Japan

Apr 2017 – Mar 2018

Research Visitor

Colorado State University, USA

Feb 2016 – Mar 2016

EDUCATION

Doctor of Engineering (PhD)

The University of Tokyo, Japan

Adviser: Prof. Shinichi Yamashita and Prof. Mitsuru Uesaka

Dissertation: *Study on Fundamental Process of Radiation Protection by Antioxidant Rutin.* (In Japanese)

Apr 2017 – Mar 2020

Master of Engineering (MEng)

The University of Tokyo, Japan

Adviser: Prof. Shinichi Yamashita and Prof. Mitsuru Uesaka

Apr 2015 – Mar 2017

Bachelor of Engineering (BE)

Chiba University, Japan

Adviser: Prof. Wenwei Yu

Apr 2010 – Mar 2015

GRANTS & FELLOWSHIPS

Grant-in-Aid for JSPS Fellows

JSPS, Japan

18J12179; 1,900,000 JPY.

Apr 2018 – Mar 2020

PEER-REVIEWED PUBLICATIONS

Published or in press

† Corresponding author

1. **Hao Yu**[†], Jackson King, Thejaswini Basappa, Sylwia Ptasińska, “Energy-dependent selective bond cleavage induced by low-energy electrons in DNA films deposited from buffered solutions: Insights from XPS”. *The Journal of Chemical Physics*, 2026, 164(8), 085104.
doi:10.1063/5.0293271
2. **Hao Yu**[†], Jackson King, Sylwia Ptasińska, “Surface chemical changes of polytetrafluoroethylene under sub-keV (10–500 eV) electron irradiation: An X-ray photoelectron spectroscopy analysis”. *Applied Surface Science*, 2026, 727, 165988.
doi:10.1016/j.apsusc.2026.165988
3. João Pereira-da-Silva, **Hao Yu**[†], Bo-An Chen, Pitambar Sapkota, Filipe Ferreira da Silva, Sylwia Ptasińska, “Interaction of 25 eV electrons with DNA constituents: XPS analysis of calf thymus DNA, nucleosides, and nucleobases”. *The European Physical Journal D*, 2025, 79(10), 120.
doi:10.1140/epjd/s10053-025-01068-w
4. **Hao Yu**, Akira Idesaki, Kimio Yoshimura, Yasunari Maekawa, “Cooperative Effects of Ni Catalyst and Radiation on Thermal Decomposition Behavior of Poly(tetrafluoroethylene)”. *ACS Omega*, 2025, 10(27), 28848–28856.
doi:10.1021/acsomega.4c11623
5. **Hao Yu**[†], Akira Idesaki, Kimio Yoshimura, Yue Zhao, Yasunari Maekawa, “Effects of temperature on the decomposition of PTFE induced by electron beam irradiation”. *Radiation Physics and Chemistry*, 2025, 237, 113029.
doi:10.1016/j.radphyschem.2025.113029
6. Hanqin Weng, Baoying Zhang, Yizhi Chen, Shinichi Yamashita, Yusa Muroya, **Hao Yu**, Xinrui Zhang, Xiaofan Yang, Weiqun Shi, “Radiolysis of diamide phenanthroline in 1-octanol: From the insights into transient processes to a versatile strategy to enhance the radiation-resistance”. *Radiation Physics and Chemistry*, 2025, 235, 112874.
doi:10.1016/j.radphyschem.2025.112874

7. Debabrata Maiti, **Hao Yu**, Jun Su An, Shinichi Yamashita, Mitsuru Naito, Kanjiro Miyata, Hyun Jin Kim, "Dual Porphyrin-Loaded Scintillating Nanoparticles Enhanced Photodynamic Therapy in Hypoxic Cancer Cells under X-ray Irradiation". *ChemBioChem*, 2025, 26(4), e202400838.
doi:10.1002/cbic.202400838
8. Chaozhong Tian, Shinichi Yamashita, Atsushi Kimura, Yui Obata, **Hao Yu**, Mitsumasa Taguchi, "Hydroxyl radical scavenging and chemical repair capabilities of positively charged peptides (PCPs): a pulse radiolysis study". *Free Radical Research*, 2024, 58(6-7), 388–395.
doi:10.1080/10715762.2024.2385342
9. **Hao Yu**, Akira Idesaki, Akihiro Hiroki, Shin Hasegawa, Kouichi Hirota, Yasunari Maekawa, "Effect of high-dose electron beam irradiation on thermal decomposition behavior of polytetrafluoroethylene (PTFE)". *Radiation Physics and Chemistry*, 2024, 216, 111435.
doi:10.1016/j.radphyschem.2023.111435
10. Debabrata Maiti, **Hao Yu**, Yuki Mochida, Seongyeon Won, Shinichi Yamashita, Mitsuru Naito, Kanjiro Miyata, Hyun Jin Kim, "Terbium-Rose Bengal Coordination Nanocrystals-Induced ROS Production under Low-Dose X-rays in Cultured Cancer Cells for Photodynamic Therapy". *ACS Applied Bio Materials*, 2023, 6(6), 2505–2513.
doi:10.1021/acsabm.3c00304
11. Atsushi Kimura, Tadashi Arai, Miho Ueno, Kotaro Oyama, **Hao Yu**, Shinichi Yamashita, Yudai Otome, Mitsumasa Taguchi, "Synthesis of Small Peptide Nanogels Using Radiation Crosslinking as a Platform for Nano-Imaging Agents for Pancreatic Cancer Diagnosis". *Pharmaceutics*, 2022, 14(11), 2400.
doi:10.3390/pharmaceutics14112400
12. Debabrata Maiti, **Hao Yu**, Beob Soo Kim, Mitsuru Naito, Shinichi Yamashita, Hyun Jin Kim, Kanjiro Miyata, "Rose Bengal Decorated NaYF₄:Tb Nanoparticles for Low Dose X-ray-Induced Photodynamic Therapy in Cancer Cells". *ACS Applied Bio Materials*, 2022, 5(11), 5477–5486.
doi:10.1021/acsabm.2c00801
13. **Hao Yu**, Yusuke Kondo, Kentaro Fujii, Akinari Yokoya, Shinichi Yamashita, "Establishment of a Method for Investigating Direct and Indirect Actions of Ionizing Radiation Using Scavenger-free Plasmid DNA". *Radiation Research*, 2022, 197(6), 594–604.
doi:10.1667/RADE-21-00057.1
14. **Hao Yu**, Shinichi Yamashita, "Radical scavenging and chemical repair of rutin observed by pulse radiolysis: as a basis for radiation protection". *Free Radical Research*, 2019, 53(9-10), 1005–1013.
doi:10.1080/10715762.2019.1667991

15. Ian M. Cartwright, Cathy Su, Jeremy S. Haskins, Victoria A. Salinas, Shigeaki Sunada, **Hao Yu**, Mitsuru Uesaka, Hirokazu Hirakawa, David J. Chen, Akira Fujimori, Takamitsu A. Kato, “DNA Repair Deficient Chinese Hamster Ovary Cells Exhibiting Differential Sensitivity to Charged Particle Radiation under Aerobic and Hypoxic Conditions”. *International Journal of Molecular Sciences*, 2018, 19(8), 2228.
doi:10.3390/ijms19082228
16. **Hao Yu**, Jeremy S. Haskins, Cathy Su, Allison Allum, Alexis H. Haskins, Victoria A. Salinas, Shigeaki Sunada, Taisuke Inoue, Yasushi Aizawa, Mitsuru Uesaka, Takamitsu A. Kato, “In vitro screening of radioprotective properties in the novel glucosylated flavonoids”. *International Journal of Molecular Medicine*, 2016, 38(5), 1525–1530.
doi:10.3892/ijmm.2016.2764

Manuscripts under review

- **Hao Yu**[†], Cecilia Julieta Garcia Villavicencio, Sylwia Ptasińska, “Quantification of Multiple DNA Damage Types in Additive-Free Aqueous Plasmid DNA under Helium Atmospheric-Pressure Plasma Jet Irradiation”. *Plasma Chemistry and Plasma Processing*.
Preprint: doi:10.21203/rs.3.rs-9645581/v1
- **Hao Yu**[†], Cecilia Julieta Garcia Villavicencio, Sylwia Ptasińska, “Modulating Effects of Amino Acids on Low-Temperature Plasma-Induced DNA Damage”. *Plasma Chemistry and Plasma Processing*.
Preprint: doi:10.21203/rs.3.rs-11067747/v1

Manuscripts in preparation

- Cecilia Julieta Garcia Villavicencio, **Hao Yu**, Luke Petruconis, Sylwia Ptasińska, “Quantification and Correlation of DNA Damage and Reactive Oxygen Species Generated by a Helium Atmospheric Pressure Plasma Jet.”

PATENTS

- Yasunari Maekawa, Akira Idesaki, **Hao Yu**, Akihiro Hiroki, Koichi Hirota, Shin Hasegawa, “Method and apparatus for decomposition of polymeric materials, method and apparatus for production of constituent unit compounds of polymeric materials.” Japanese Patent Application No. 2024-114780, July 18, 2024, claiming domestic priority from No. 2023-117122; published as JP 2025-015497 A.

BOOK CHAPTERS

- **Hao Yu**[†], Akira Idesaki, Yasunari Maekawa, “High-dose Radiation-induced Decomposition of Teflon (PTFE).” *PFAS Regulatory Trends and Corresponding Technologies*, Technical Information Institute, June 2024. ISBN: 978-4-86798-027-9. (In Japanese)

SOCIETY REVIEW ARTICLES

- Chaozhong Tian, Yui Obata, Shinichi Yamashita, **Hao Yu**, Atsushi Kimura, Mitsumasa Taguchi, “Chemical Repair Capabilities of Positively Charged Peptides Containing Lysine and Histidine.” *Radiation Chemistry*, No. 119, p. 23 (2025).
- Baoying Zhang, Yizhi Chen, Hanqin Weng, Shinichi Yamashita, Yusa Muroya, **Hao Yu**, Xinrui Zhang, Xiaofan Yang, Weiqun Shi, “A new strategy to enhance the radiation stability of diamide phenanthroline extractants: starting from the key elementary reactions of radiolysis”. *Journal of Radiation Research and Radiation Processing*, 2025, 43(5), 051002.
doi:10.11889/j.1000-3436.2025-0065 (In Chinese)
- Chaozhong Tian, Yui Obata, Shinichi Yamashita, Atsushi Kimura, **Hao Yu**, Mitsumasa Taguchi, “Effect of Histidine in Positively Charged Peptides (PCPs) on Free Radical Scavenging and Chemical Repair.” *Radiation Chemistry*, No. 117, p. 23 (2024).
- Miho Ueno, Tadashi Arai, **Hao Yu**, Shinichi Yamashita, Atsushi Kimura, Mitsumasa Taguchi, “Development of Radiation Crosslinked Peptide Nanoparticles for Drug Discovery Application: 1. Analysis of Radiation Chemical Reactions.” *Radiation Chemistry*, No. 113, p. 71 (2022). (In Japanese)
- **Hao Yu**[†], Yusuke Kondo, Kentaro Fujii, Akinari Yokoya, Shinichi Yamashita, “Effect of Water Content on Radiation-Induced DNA Damage: 2. Base Damage and AP Sites.” *Radiation Chemistry*, No. 111, p. 63 (2021). (In Japanese)
- **Hao Yu**[†], Shinichi Yamashita, Mitsuru Uesaka, “Radioprotective Effects of Antioxidant MG-Rutin: Observation of Radical Scavenging and Chemical Recovery by Pulse Radiolysis.” *Radiation Chemistry*, No. 105, p. 57 (2018). (In Japanese)

PRESENTATIONS

Invited

- **Hao Yu**, Kentaro Fujii, Akinari Yokoya, Shinichi Yamashita, “Chemical Repair Effects of Additives on Radiation-Induced DNA Initial Damage.” *64th Annual Meeting of the Japanese Radiation Research Society*, held online, September 2021.

Oral

- **Hao Yu**, Cecilia Julieta Garcia Villavicencio, Sylwia Ptasińska, “Separating Radical-Driven and pH-Dependent Pathways in Low-Temperature Plasma-Induced DNA Damage.” Accepted contributed oral presentation, *AVS 72nd International Symposium & Exhibition*, Pittsburgh, PA, USA, scheduled November 13, 2026 (program PS2-FrM-11).
- **Hao Yu**, João Pereira-da-Silva, Jackson King, Sylwia Ptasińska, “XPS Analysis of DNA Damage Induced by Low-Energy Electrons (1–25 eV): Site and Energy Dependence.” *Midwest Regional Meeting of the American Chemical Society (MWRM 2025)*, Columbia, MO, USA, October 2025.

- **Hao Yu**, João Pereira-da-Silva, Jackson King, Thejaswini Basappa, Bo-An Chen, Pitambar Sapkota, Filipe Ferreira da Silva, Sylwia Ptasińska, “Application of XPS to Detect Low-Energy Electron (1–25 eV) Irradiation-Induced DNA Degradation.” *AVS 71st International Symposium and Exhibition*, Charlotte, NC, USA, September 2025.
- **Hao Yu**, Akira Idesaki, Kimio Yoshimura, Shin Hasegawa, Yue Zhao, Yasunari Maekawa, “Synergistic effects of temperature, atmosphere, and catalyst on irradiation-induced PTFE decomposition.” *4th International Conference on Ionizing Processes (ICIP 2024)*, Notre Dame, IN, USA, August 2024.
- **Hao Yu**, Akira Idesaki, Kimio Yoshimura, Akihiro Hiroki, Yasunari Maekawa, “Effects of Nickel on the Acceleration of Radiation-induced Decomposition of Polytetrafluoroethylene (PTFE).” *2024 MRS Spring Meeting and Exhibit*, Seattle, WA, USA, April 2024.
- **Hao Yu**, Akira Idesaki, Akihiro Hiroki, Shin Hasegawa, Koichi Hirota, Yasunari Maekawa, “Decrease of Decomposition Temperature of PTFE Powder by Electron Beam Irradiation and Additives.” *3rd Materials Research Meeting and 24th IUMRS International Conference in Asia (MRM2023/IUMRS-ICA2023)*, Kyoto, Japan, December 2023.
- **Hao Yu**, Kentaro Fujii, Akinari Yokoya, Shinichi Yamashita, “Radiation-Induced DNA Damage and Selective Protection by Antioxidants.” *Miller Online Workshop on Radiation Chemistry*, held online, February 2022.
- **Hao Yu**, Kentaro Fujii, Akinari Yokoya, Shinichi Yamashita, “Effects of Additives on Radiation-Induced DNA Damage: From the Viewpoints of Free Radical Scavenging and Chemical Repair.” *31st Miller Conference on Radiation Chemistry*, Workington, UK, September 2019.
- **Hao Yu**, Shinichi Yamashita, “Radical Scavenging and Chemical Repair of a Natural Flavonoid Rutin: A Pulse Radiolysis Study.” *7th Asia-Pacific Symposium on Radiation Chemistry (APSRC-2018)*, Shanghai, China, November 2018.

Posters

- **Hao Yu**, Jackson King, Akira Idesaki, Yasunari Maekawa, Sylwia Ptasińska, “Energy Dependent Electron Induced Modification of PTFE: Bulk Degradation and Surface Defluorination.” *Annual Meeting of the Council on Ionizing Radiation Measurements and Standards (CIRMS 2026)*, Washington, DC, USA, April 2026.
- **Hao Yu**, Jackson King, Sylwia Ptasińska, “XPS Analysis of PTFE Decomposition Induced by 10–500 eV Electron Beam Irradiation.” *AVS 71st International Symposium and Exhibition*, Charlotte, NC, USA, September 2025.
- **Hao Yu**, Akira Idesaki, Kimio Yoshimura, Yasunari Maekawa, “Gaseous Product Analysis in Irradiation-Induced Decomposition of Perfluorocarbons.” *4th International Conference on Ionizing Processes (ICIP 2024)*, Notre Dame, IN, USA, August 2024.
- **Hao Yu**, Akira Idesaki, Akihiro Hiroki, Shin Hasegawa, Koichi Hirota, Kimio Yoshimura, Yue Zhao, Yasunari Maekawa, “Quantum Beams for Efficiently Promoting PTFE Decomposition.” *The 7th QST International Symposium*, July 2024.

- **Hao Yu**, Akira Idesaki, Akihiro Hiroki, Shin Hasegawa, Koichi Hirota, Yasunari Maekawa, "Effect of Electron Beam Irradiation on Thermal Decomposition Behavior of Polytetrafluoroethylene (PTFE)." *The 21st International Conference on Radiation Effects in Insulators (REI-21)*, September 2023.

SERVICE & PEER REVIEW

- Reviewer, *Polymer Degradation and Stability* (Elsevier), 2026.

AWARDS & HONORS

- R&D Achievement Award, Takasaki Institute for Advanced Quantum Science (QST), Japan, March 2026.
- Excellent Poster Presentation Award, QST Takasaki Science Festival 2023, December 2023.
- Excellent Presentation Award for Young Researchers, 63rd Symposium on Radiation Chemistry, Japan Society of Radiation Chemistry, December 2020.
- Excellent Poster Presentation Award, Summer School for Young Radiation Chemists, Young Researchers' Association of the Japan Society of Radiation Chemistry, October 2019.
- Excellent Presentation Award for Young Researchers, 55th Isotope and Radiation Research Symposium, Japan Radioisotope Association, July 2018.
- Excellent Presentation Award for Young Researchers, 60th Symposium on Radiation Chemistry, Japan Society of Radiation Chemistry, September 2017.
- Excellent Presentation Award, Student Division, The Young Researcher Presentation Meeting, North Kanto Branch, Atomic Energy Society of Japan, April 2017.
- Encouragement Presentation Award, 10th Student Research Presentation Meeting of the Kanto-Koshinetsu Branch, Atomic Energy Society of Japan, March 2017.
- Best Poster Presentation Award, Summer School for Young Radiation Chemists, Young Researchers' Association of the Japan Society of Radiation Chemistry, October 2016.